SRI SRINIVASA EDUCATIONAL & CHARITABLE TRUST (R)
SAPTHAGIRI COLLEGE OF ENGINEERING
(Affiliated to Visvesvaraya Technological University, Belagavi and Approved by AICTE-New Delhi)
(Accredited by NAAC "A" Grade & NBA) (ISO 9001:2015 & ISO 14001:2015 Certified)
#14/5, Chikkasandra, Hesaraghatta Main Road, Bengaluru-560057

DEPARTMENT OF MATHEMATICS

ACADEMIC YEAR: 2023-24 EVEN SEMESTER (BMAE201)

QUESTION BANK

MODULE 2

1. Prove that the subset W={(x,y, z)|x–3y+4z =0}of the vector space R^3 is subspace of R^3.

2. Using the convolution theorem, find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$.

3. Prove that $T:R^3 \rightarrow R^3$ be defined by
$T\,(a,b,c)=(3a,a-b,2a+b+c)$ is a linear transformation.

4. Find the Laplace transform of (i) $e^{-4t}(2cos6t - 3sin5t)$ (ii) $\frac{(\cos 2t - \cos 3t)}{t}$

5. Find f (9) from the data by Newton's divided difference formula:

x	5	7	11	13	17
F(x)	150	392	1452	2366	5202

6. Use Lagrange's interpolation formula to find y at $x = 4$, for the given data:

X	0	1	2	5
y	2	3	12	147

7. Using Modified Euler's Method find y at $x = 0.1$ given $y' = x^2 + y$ with $y(0) = 1, \quad h = 0.05$

8. Apply Milne's predictor-corrector formulae to compute $y(4.5)$ given that $5x\frac{dy}{dx} = 2 - y^2$ and

X	4.1	4.2	4.3	4.4
y	1.0049	1.0097	1.0143	1.0187

9. Find $div\ \vec{F}$ and $curl\ \vec{F}$ at (1,2,3), if $\vec{F} = 3x^2\hat{\imath} + 5xy^2\hat{\jmath} + 5xyz^3\hat{k}$

10. Show that $\vec{A} = (y^2 - z^2 + 3yz - 2x)\hat{\imath} + (3xz + 2xy)\hat{\jmath} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational